

6.8 From Eq. (6.100), we know that a rotating polarization gives rise to an effective magnetization,

with $\vec{M} = \vec{P} \times \vec{v}$, where

$$\vec{P} = 3\epsilon_0 \epsilon_0 \left(\frac{\epsilon - \epsilon_0}{\epsilon + 2\epsilon_0} \right) \hat{x}, \quad \text{and} \quad \vec{v} = \vec{\omega} \times \vec{r} = \omega r (\hat{z} \times \hat{r}).$$

$$\begin{aligned} \text{Then, } \vec{M} &= \vec{P} \times \vec{v} = 3\epsilon_0 \epsilon_0 \omega r \left(\frac{\epsilon - \epsilon_0}{\epsilon + 2\epsilon_0} \right) [\hat{x} \times (\hat{z} \times \hat{r})] \\ &= 3\epsilon_0 \epsilon_0 \omega r \left(\frac{\epsilon - \epsilon_0}{\epsilon + 2\epsilon_0} \right) (\hat{z}(\hat{x} \cdot \hat{r}) - \hat{r}(\hat{x} \cdot \hat{z})) \\ &= 3\epsilon_0 \epsilon_0 \omega \left(\frac{\epsilon - \epsilon_0}{\epsilon + 2\epsilon_0} \right) [r \sin\theta \cos\phi \hat{r} - r \sin^2\theta \cos\phi \hat{\theta}], \end{aligned}$$

Where we have used the fact that $\hat{x} \cdot \hat{z} = 0$ and $\hat{z} = \cos\theta \hat{r} - \sin\theta \hat{\theta}$ in spherical coordinates. It is straightforward to verify that

$$\nabla \cdot \vec{M} = 3\epsilon_0 \epsilon_0 \omega \left(\frac{\epsilon - \epsilon_0}{\epsilon + 2\epsilon_0} \right) \left[\frac{1}{r^2} \frac{\partial}{\partial r} (r^3 \sin\theta \cos\phi) - \frac{1}{r \sin\theta} \frac{\partial}{\partial \theta} (r \sin^3\theta \cos\phi) \right] = 0.$$

Thus, by Eq. (5.100), the contribution to the magnetic scalar potential comes only from surface magnetic charge density,

$$\Phi_M = \frac{1}{4\pi} 3\epsilon_0 \epsilon_0 \omega \left(\frac{\epsilon - \epsilon_0}{\epsilon + 2\epsilon_0} \right) \oint \frac{a \sin\theta' \cos\theta' \cos\phi'}{|\vec{x} - \vec{x}'|} da'.$$

Using the expansion for $1/|\vec{x} - \vec{x}'|$,

$$\frac{1}{|\vec{x} - \vec{x}'|} = 4\pi \sum_{l=0}^{\infty} \sum_{m=-l}^l \frac{1}{2l+1} \frac{r_<^l}{r_>^{l+1}} Y_{lm}(\theta, \phi) Y_{lm}^*(\theta', \phi'),$$

and also notice that

$$\sin\theta' \cos\theta' \cos\phi' = -\frac{1}{2} \sqrt{\frac{8\pi}{15}} (Y_{21}(\theta', \phi') - Y_{2,-1}(\theta', \phi')),$$

the surface integral becomes

$$\begin{aligned} \oint \frac{a \sin\theta' \cos\theta' \cos\phi'}{|\vec{x} - \vec{x}'|} da' &= 4\pi a^3 \sum_{l=0}^{\infty} \sum_{m=-l}^l \frac{1}{2l+1} \frac{r_<^l}{r_>^{l+1}} Y_{lm}(\theta, \phi) \int -\frac{1}{2} \sqrt{\frac{8\pi}{15}} (Y_{21}(\theta', \phi') - Y_{2,-1}(\theta', \phi')) Y_{lm}^*(\theta', \phi') da' \\ &= 4\pi a^3 \sum_{l=0}^{\infty} \sum_{m=-l}^l \frac{1}{2l+1} \frac{r_<^l}{r_>^{l+1}} Y_{lm}(\theta, \phi) \left[-\frac{1}{2} \sqrt{\frac{8\pi}{15}} (\delta_{l,2} \delta_{m,1} - \delta_{l,2} \delta_{m,-1}) \right] \end{aligned}$$

$$= 4\pi a^3 \frac{1}{5} \frac{r_c^2}{r_s^3} \left[-\frac{1}{2} \sqrt{\frac{8\pi}{15}} \left(Y_{21}(0, \phi) - Y_{2,-1}(0, \phi) \right) \right] = 4\pi a^3 \frac{1}{5} \frac{r_c^2}{r_s^3} \sin\theta \cos\phi$$

Therefore,

$$\Phi_m = \frac{1}{4\pi} 3 \epsilon_0 \bar{E}_0 \omega \left(\frac{\epsilon - \epsilon_0}{\epsilon + 2\epsilon_0} \right) \cdot 4\pi a^3 \frac{1}{5} \frac{r_c^2}{r_s^3} \sin\theta \cos\phi$$

$$= \frac{3}{5} \left(\frac{\epsilon - \epsilon_0}{\epsilon + 2\epsilon_0} \right) \epsilon_0 \bar{E}_0 \omega \frac{a^3}{r_s^3} \frac{r_c^2}{r^2} r \sin\theta \cos\phi r \cos\theta$$

$$= \frac{3}{5} \left(\frac{\epsilon - \epsilon_0}{\epsilon + 2\epsilon_0} \right) \epsilon_0 \bar{E}_0 \omega \frac{a^3}{r_s^3} \frac{r_c^2}{r^2} \pi z$$

If $r > a$, then $r_s = r$, $r_c = a$, and $\frac{a^3}{r_s^3} \frac{r_c^2}{r^2} = \left(\frac{a}{r_s} \right)^5$ On the other hand, if $r < a$, then

$\frac{a^3}{r_s^3} \frac{r_c^2}{r^2} = 1$, which can still be expressed as $(a/r_s)^5$.

Finally,
$$\Phi_m = \frac{3}{5} \left(\frac{\epsilon - \epsilon_0}{\epsilon + 2\epsilon_0} \right) \epsilon_0 \bar{E}_0 \omega \left(\frac{a}{r_s} \right)^5 \pi z$$